

ASYMPTOTICALLY MATCHED EXTRAPOLATION OF FISHNET FAILURE PROBABILITY TO CONTINUUM SCALE

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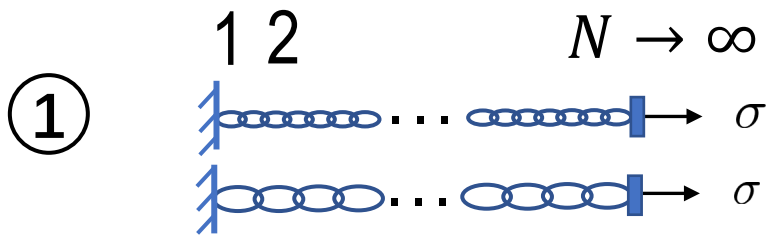
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Analytically Tractable Strength Probability Models

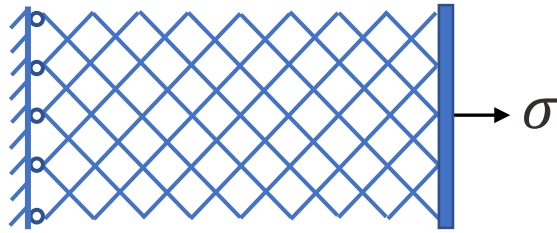
Until 2017, there existed only two probabilistic models of structural strength

- The series coupling model, a.k.a. the weakest-link model, which led in 1939 to the Weibull distribution at the large size limit (derived earlier by Fisher and Tippett in 1928 without reference to material strength, as one of only three possible extreme value distributions).
- The parallel coupling model, better known as the fiber bundle model, was shown in 1945 by Daniels to lead asymptotically to the Gaussian (or normal) distribution.



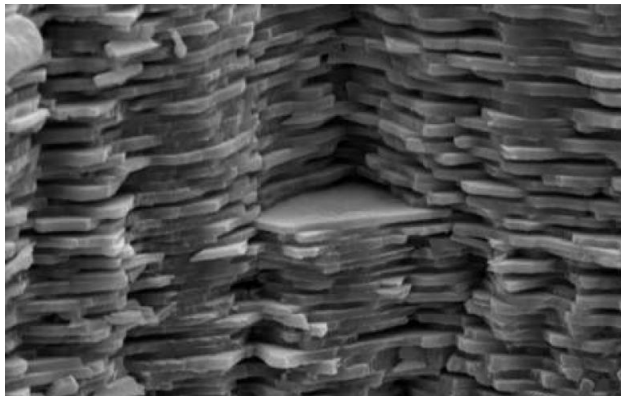
Analytically Tractable Strength Probability Models

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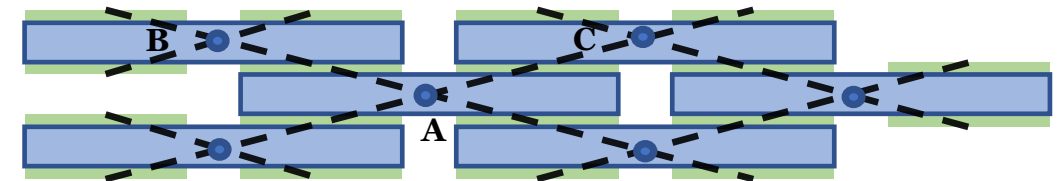
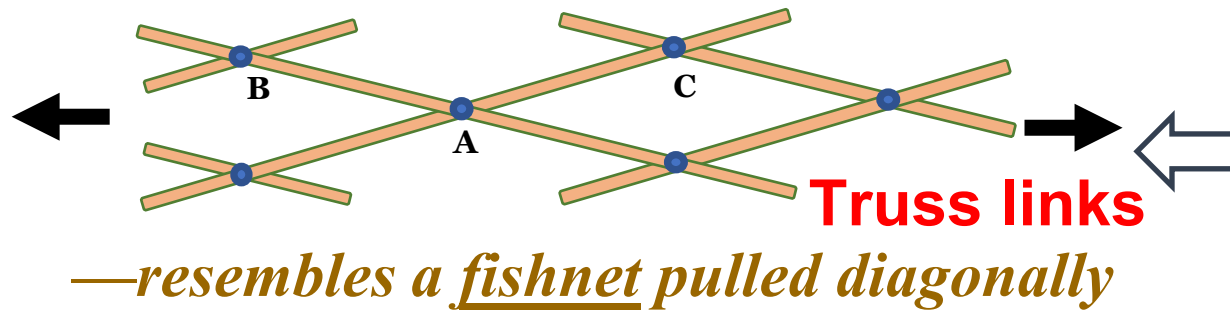
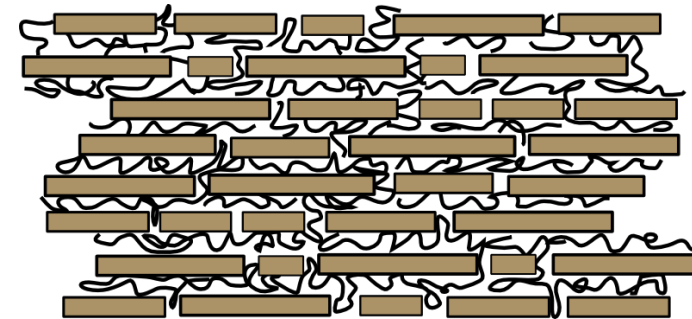
Fishnet statistics (NU 2017)

– alternating series and parallel couplings



Electron microscopy image of a fractured surface of nacre

nacre as the clue



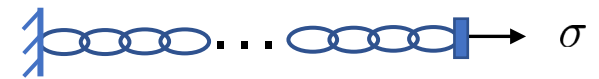
Decomposition of Fishnet Failure Probability

$$1 - P_f(\sigma) = P_{S_0}(\sigma) + P_{S_1}(\sigma) + P_{S_2}(\sigma) + \dots$$

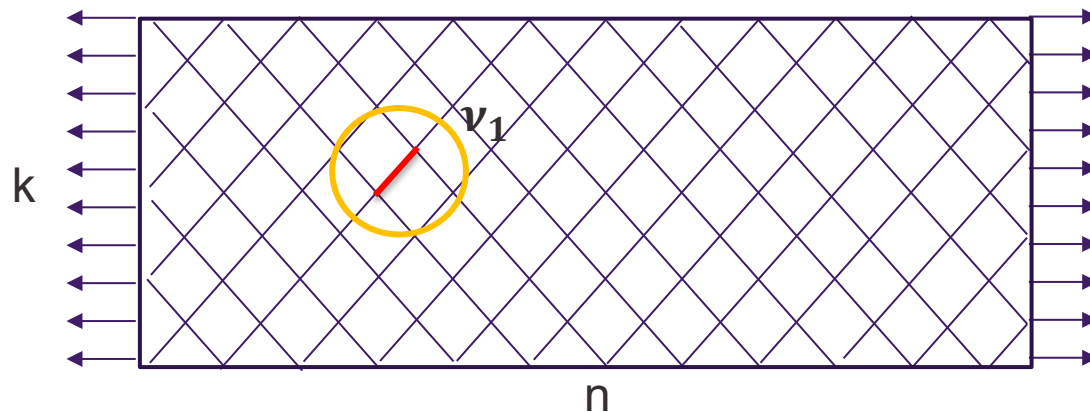
Survival probability = Survive with 0 link fail + Survive with 1 link fail + Survive with 2 link fail + ...

$$P_{S_0} = [1 - P_1(\sigma)]^N \quad \text{event } \{X(\sigma) = 0\}: \text{all } N \text{ links survive}$$

where $P_1(\sigma)$ = given failure probability of one link



$$P_{S_1}(\sigma) = N P_1(\sigma) [1 - P_1(\eta_1 \sigma)]^{v_1} [1 - P_1(\sigma)]^{N-v_1-1}$$



v_1 = number of links affected by the first failed link
 η_1 = stress concentration factor

event $\{X(\sigma) = 1\}$: only 1 link fails randomly, but all other $N - 1$ links survive

Decomposition of Fishnet Failure Probability

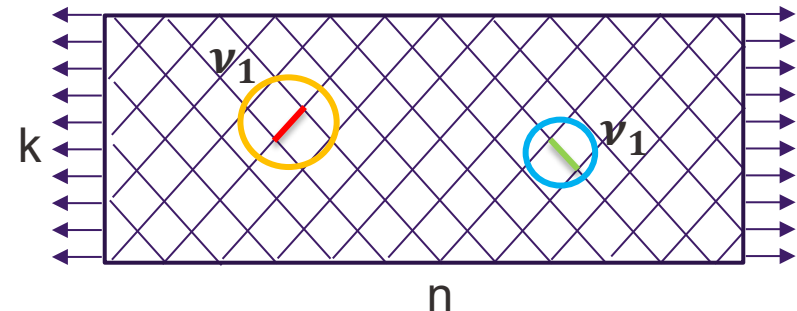
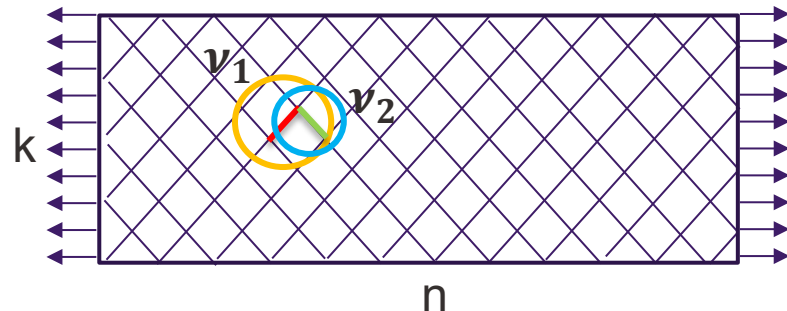
$$P_{S_2} = P_{S_{21}} + P_{S_{22}} \quad \text{event } \{X(\sigma) = 2\}: \text{ only 2 link fails randomly, but all other } N - 2 \text{ links survive}$$

$$P_{S_{21}} = N v_1 \int_0^\sigma \int_{x_1}^\sigma f(x_1) f(x_2) dx_1 dx_2 [1 - P_1(\sigma)]^{N-v_2-2} [1 - P_1(\eta_2 \sigma)]^{v_2} \\ + N v_1 \int_0^\sigma \int_\sigma^{\eta_1 \sigma} f(x_1) f(x_2) dx_1 dx_2 [1 - P_1(\eta_2 \sigma)]^{v_2}$$

strengths of both failed links are smaller than σ

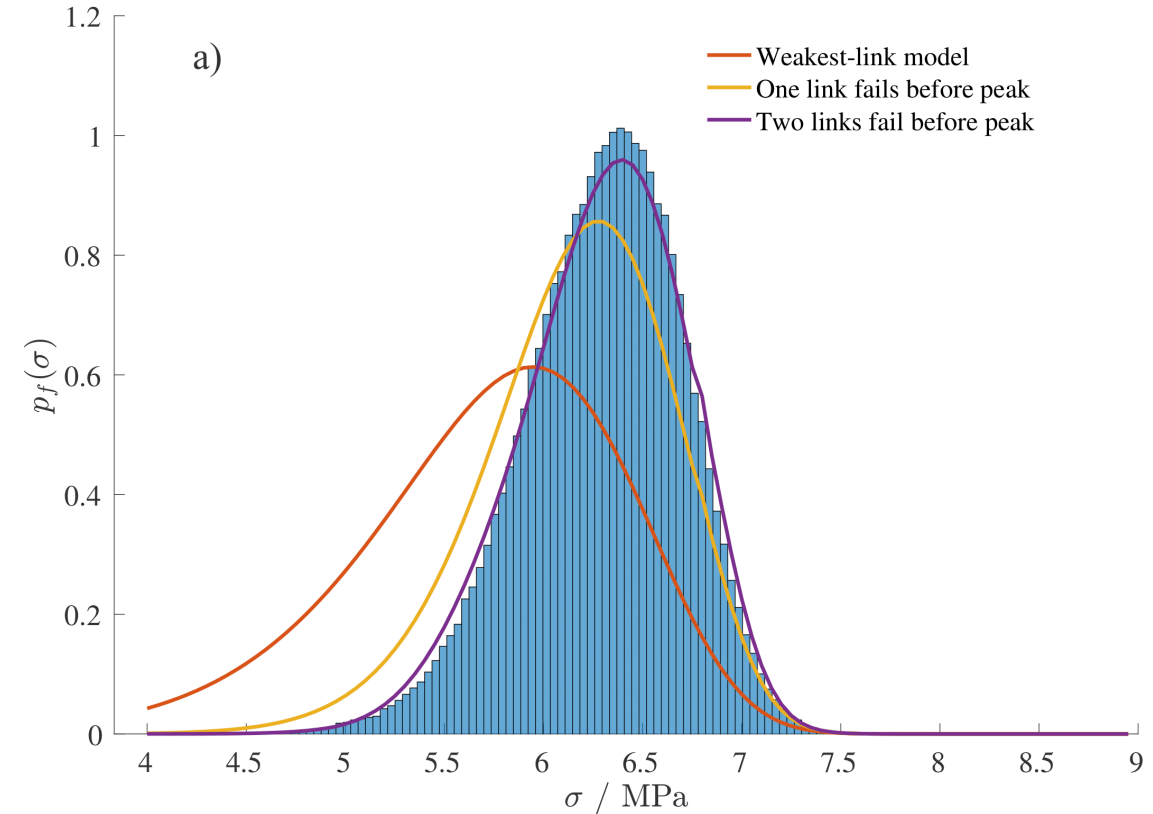
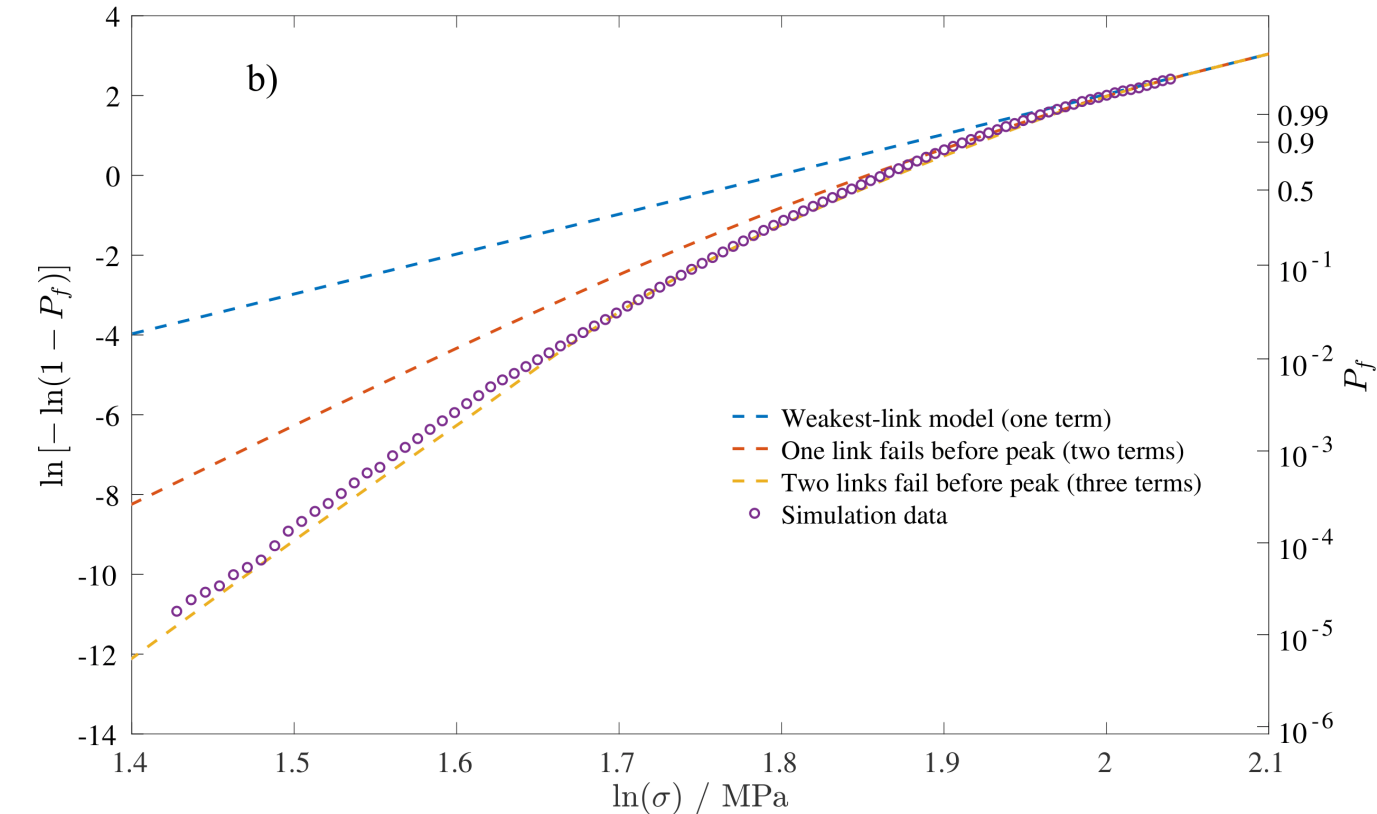
strength of the first failed link is smaller than σ and the strength of the second one is between σ and $\eta_1 \sigma$

$$P_{S_{22}} = N(N - v_1 - 1) \int_0^\sigma \int_{x_1}^\sigma f(x_1) f(x_2) dx_1 dx_2 [1 - P_1(\sigma)]^{N-2v_1-2} [1 - P_1(\eta_1 \sigma)]^{2v_1}$$



Decomposition of Fishnet Failure Probability

$$X = \ln \sigma_N, \quad Y = \ln[-\ln(1 - P_f)]$$



Crack in Fishnet Continuum (Laplace equation)

Displacement field $u(z)$: $\nabla^2 u(z) = 0$, $z = x + iy$

Stress vectors in infinite fishnet continuum with crack of length $2a$ under tension at infinity:

$$\sigma_x = -\frac{\sigma z}{\sqrt{a^2 - z^2}}, \quad \sigma_y = \frac{\sigma z}{\sqrt{z^2 - a^2}}$$

Separated near tip stress field:

$$\sigma_r = \frac{K}{\sqrt{r}} \sin \frac{\theta}{2}, \quad \sigma_\theta = \frac{K}{2\sqrt{r}} \cos \frac{\theta}{2},$$

Stress intensity factor:

$$K = \sigma \sqrt{2a} \quad \left[\frac{dK}{da} \right]_\sigma > 0$$

→ The geometry is positive

- crack become dynamic as soon as a row of links of length $2a$ forms
- This **proves** many potential random crack locations must lead to the **Weibull distribution** (i.e., the weakest link model)

Asymptotic Analysis of Fishnet Failure Probability

$$[1 - P_1(\sigma)]^N \approx 1 - NP_1(\sigma) + \frac{N(N-1)}{2} [P_1(\sigma)]^2$$

$$P_{S_1} \approx NP_1(\sigma) - N(N - \nu_1 - 1)[P_1(\sigma)]^2 - N\nu_1 P_1(\sigma)P_1(\lambda_1\sigma)$$

$$P_f = 1 - (P_{S_0} + P_{S_1}) \approx \left\{ N(N - \nu_1 - 1) - \frac{N(N-1)}{2} \right\} [P_1(\sigma)]^2 + N\nu_1 P_1(\sigma)P_1(\eta_1\sigma)$$

Extend this analysis to the three-term fishnet

$$P_f(\sigma) = \frac{N}{6} \{ N^2 + 3\nu_1^2 - 3N - 3\nu_1 + 2 \} [P_1(\sigma)]^3 - \frac{N\nu_1}{2} (\nu_1 - 1) P_1(\sigma) [P_1(\lambda_1\sigma)]^2 \\ - \frac{N\nu_1\nu_2}{2} P_1(\lambda_2\sigma) [P_1(\sigma)]^2 + N\nu_1\nu_2 P_1(\sigma) P_1(\lambda_1\sigma) P_1(\lambda_2\sigma)$$

How Is It Derived?

To determine the tail behavior of the three-term fishnet model, we can perform the same analysis except that we retain all the terms up to the order of $[P_1(\sigma)]^3$. i.e.:

$$[1 - P_1(\sigma)]^N \approx 1 - NP_1(\sigma) + \frac{N(N-1)}{2}[P_1(\sigma)]^2 - \frac{N(N-1)(N-2)}{6}[P_1(\sigma)]^3 \quad (10)$$

$$\begin{aligned} P_{s1} \approx NP_1(\sigma) - N(N - \nu_1 - 1)[P_1(\sigma)]^2 - N\nu_1 P_1(\sigma)P_1(\lambda_1\sigma) + N\nu_1(N - \nu_1 - 1)P_1(\lambda_1\sigma)[P_1(\sigma)]^2 \\ + \frac{1}{2}N(N - \nu_1 - 1)(N - \nu_1 - 2)[P_1(\sigma)]^3 + \frac{1}{2}N\nu_1(\nu_1 - 1)P_1(\sigma)[P_1(\lambda_1\sigma)]^2 \end{aligned} \quad (11)$$

$$\begin{aligned} P_{s21} \approx N\nu_1 P_1(\sigma)P_1(\lambda_1\sigma) - \frac{N\nu_1}{2}[P_1(\sigma)]^2 - \frac{1}{2}N\nu_1(N - \nu_2 - 2)[P_1(\sigma)]^3 \\ - N\nu_1\nu_2 P_1(\sigma)P_1(\lambda_1\sigma)P_1(\lambda_2\sigma) + \frac{1}{2}N\nu_1\nu_2[P_1(\sigma)]^2 P_1(\lambda_2\sigma) \end{aligned} \quad (12)$$

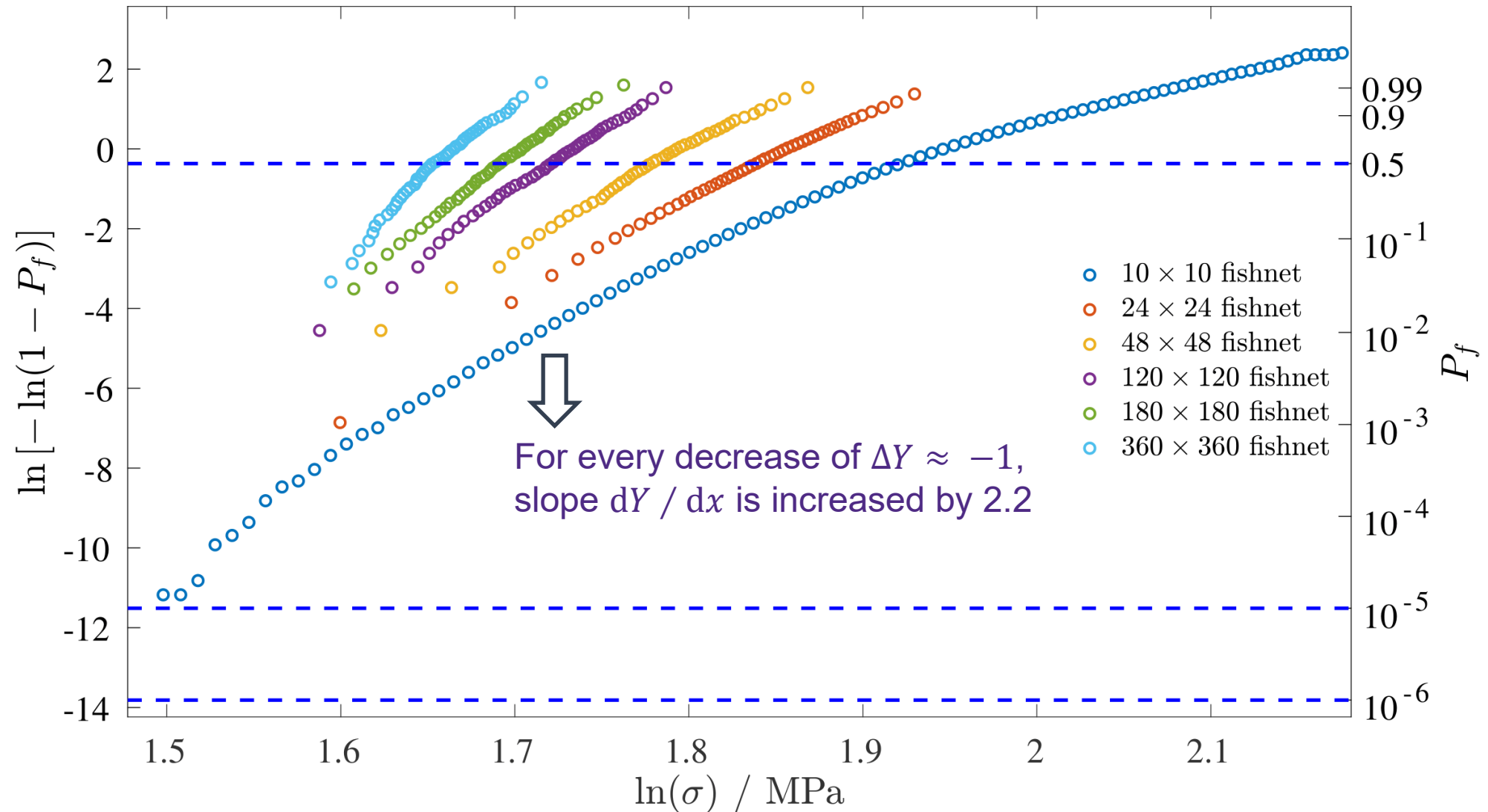
$$P_{s22} \approx \frac{1}{2}N(N - \nu_1 - 1)[P_1(\sigma)]^2 - \frac{1}{2}N(N - \nu_1 - 1)(N - 2\nu_1 - 2)[P_1(\sigma)]^3 - N(N - \nu_1 - 1)\nu_1[P_1(\sigma)]^2 P_1(\lambda_1\sigma) \quad (13)$$

Substitution of Eqs. 9- 12 into Eq. 1 yields

$$\begin{aligned} P_f(\sigma) = \frac{N}{6} \{N^2 + 3\nu_1^2 - 3N - 3\nu_1 + 2\} [P_1(\sigma)]^3 - \frac{N\nu_1}{2}(\nu_1 - 1)P_1(\sigma) [P_1(\lambda_1\sigma)]^2 \\ - \frac{N\nu_1\nu_2}{2}P_1(\lambda_2\sigma)[P_1(\sigma)]^2 + N\nu_1\nu_2 P_1(\sigma)P_1(\lambda_1\sigma)P_1(\lambda_2\sigma) \end{aligned} \quad (14)$$

Clearly, we have $P_f(\sigma) \propto \sigma^{3m}$ for small σ .

Monte Carlo Simulation of Fishnet Failure

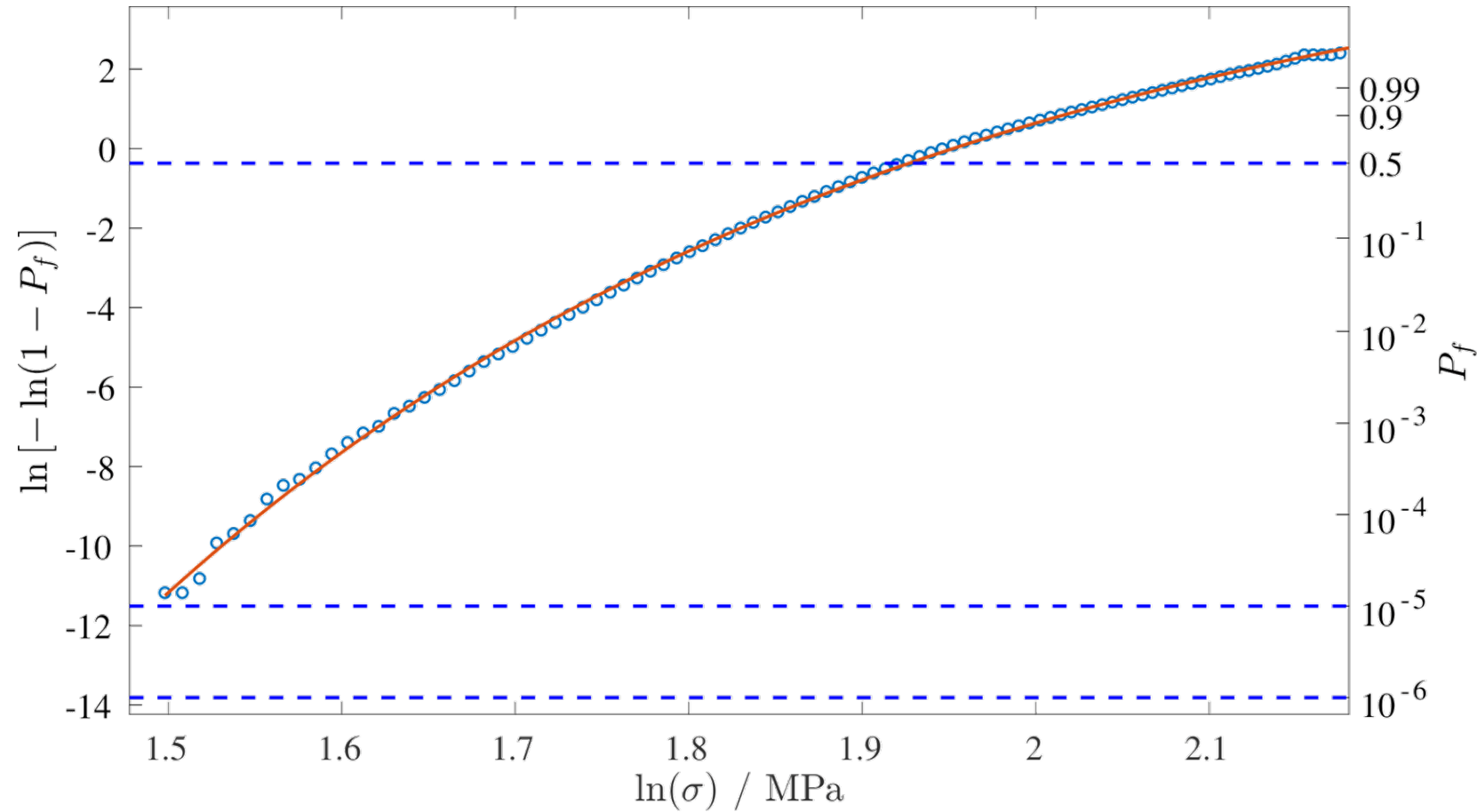


Monte Carlo Simulation of Fishnet Failure

$$\frac{dY}{dX} = m' \left(1 - \frac{Y}{p} \right), \quad X = \ln \sigma_N$$

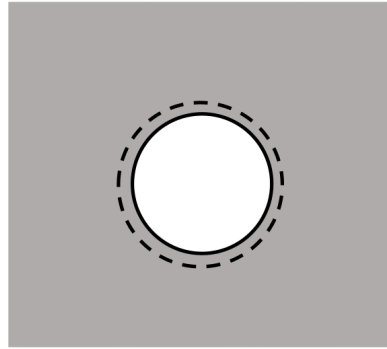


$$Y = p - \left(\frac{q}{\sigma_N} \right)^r, \quad r = \frac{m'}{p}$$



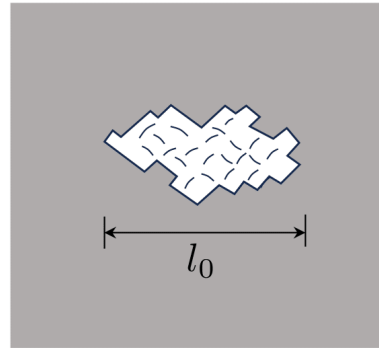
Strength Distribution of Large Fishnet

a) Unstable

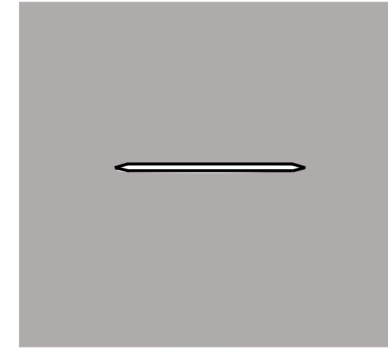


If $W > b\Gamma_{\text{surf}}$

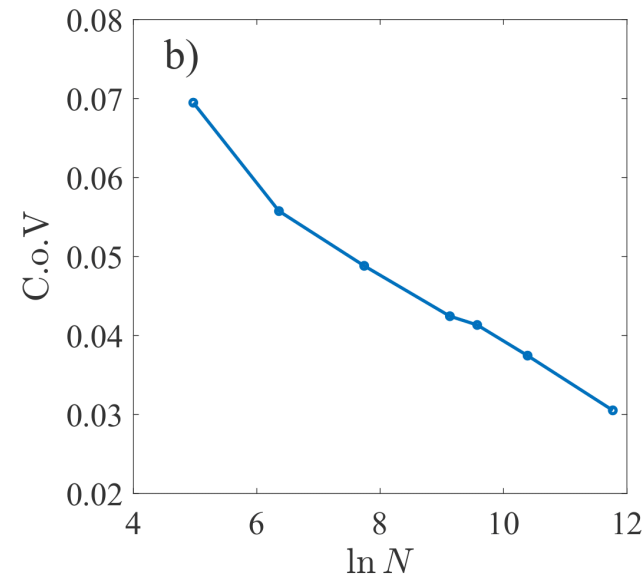
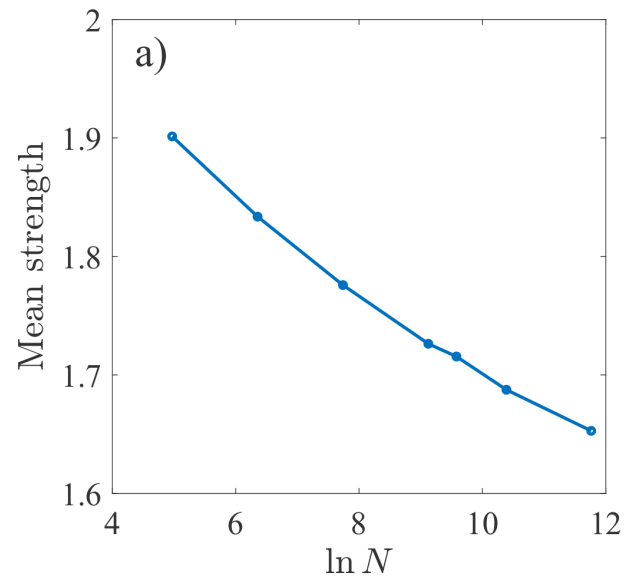
b) Inferred as unstable



c) Unstable



If $J > bG_f$



Typical Damage Patterns of Fishnets at the Peak Load

Concentration of Damage Zone:

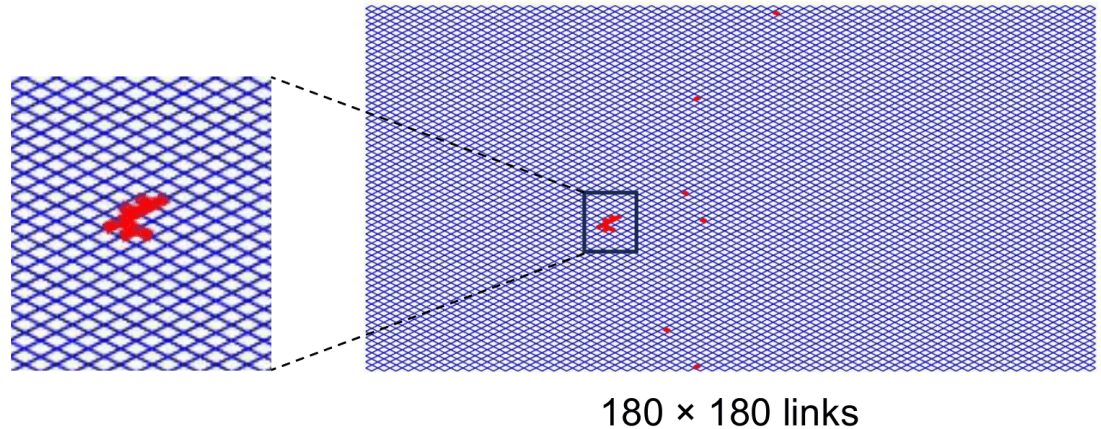
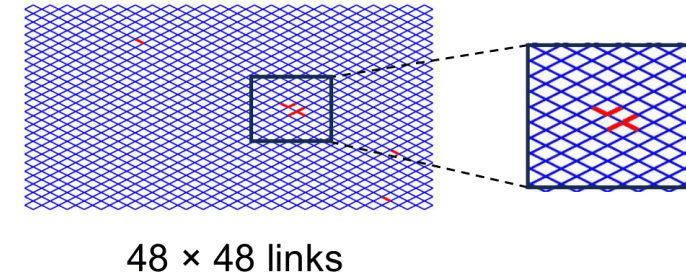
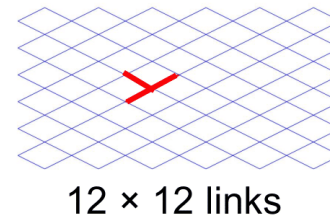
- A concentrated damage zone forms.
- Isolated failed links can also be observed in other parts.

Link Behavior during Loading:

- As loading progresses, a row of links zips through from the initial damage zone.

Size-Dependent Failure:

- The number of failed links at peak load **increases** with specimen size.

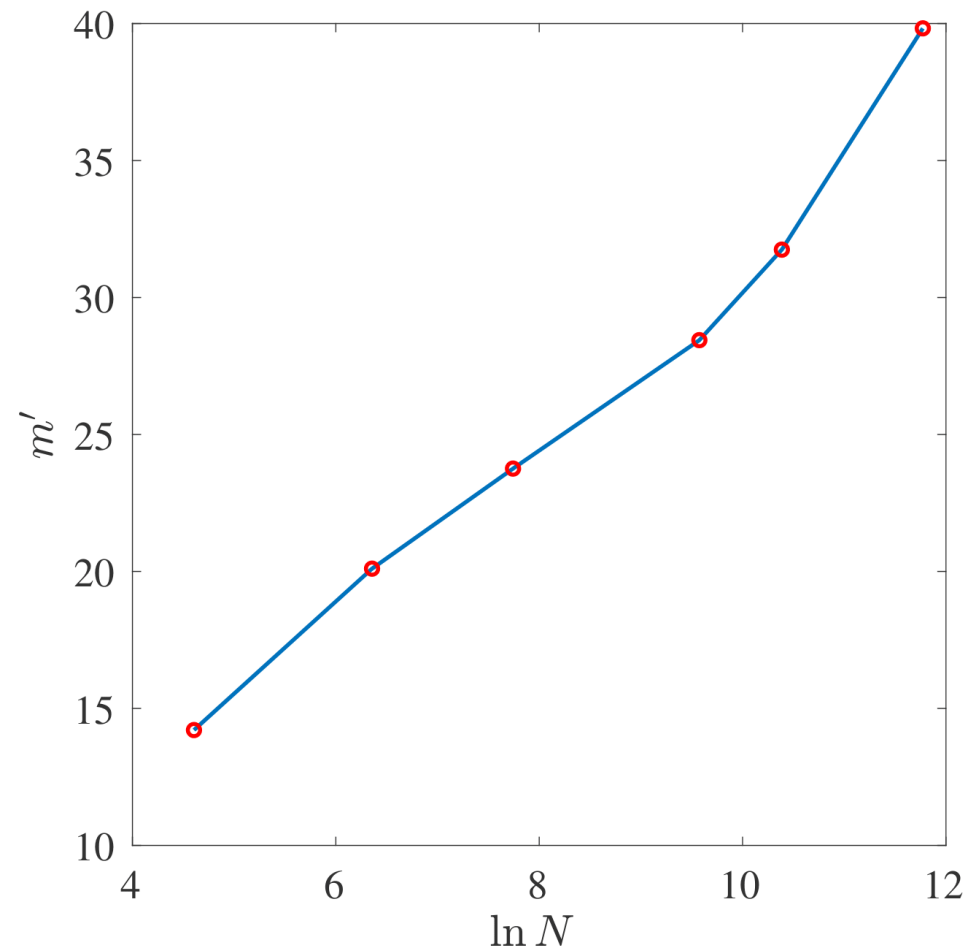
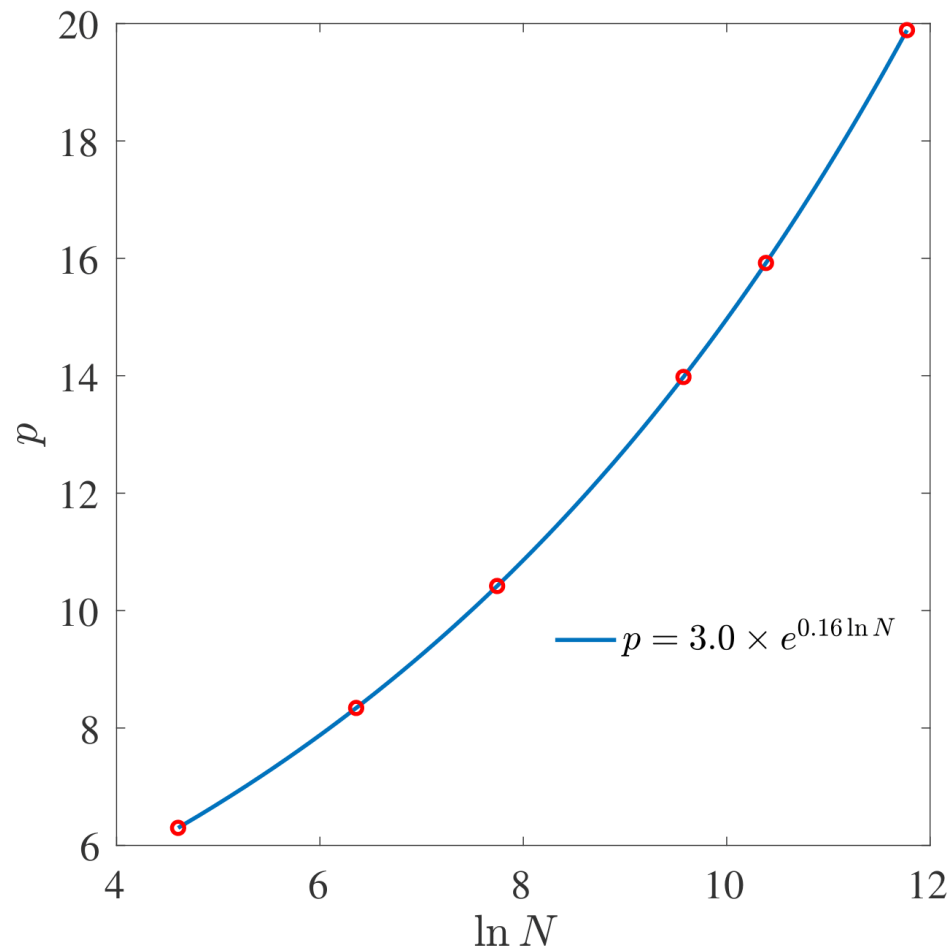


Monte Carlo Simulation of Fishnet Failure

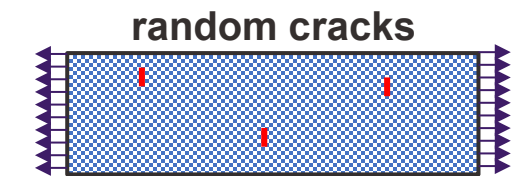
$$\frac{dY}{dX} = m' \left(1 - \frac{Y}{p} \right), \quad X = \ln \sigma_N$$

Benefits:

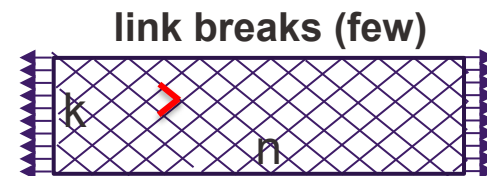
1. Enhance uniqueness
2. Reduce curvature when $N \rightarrow \infty$



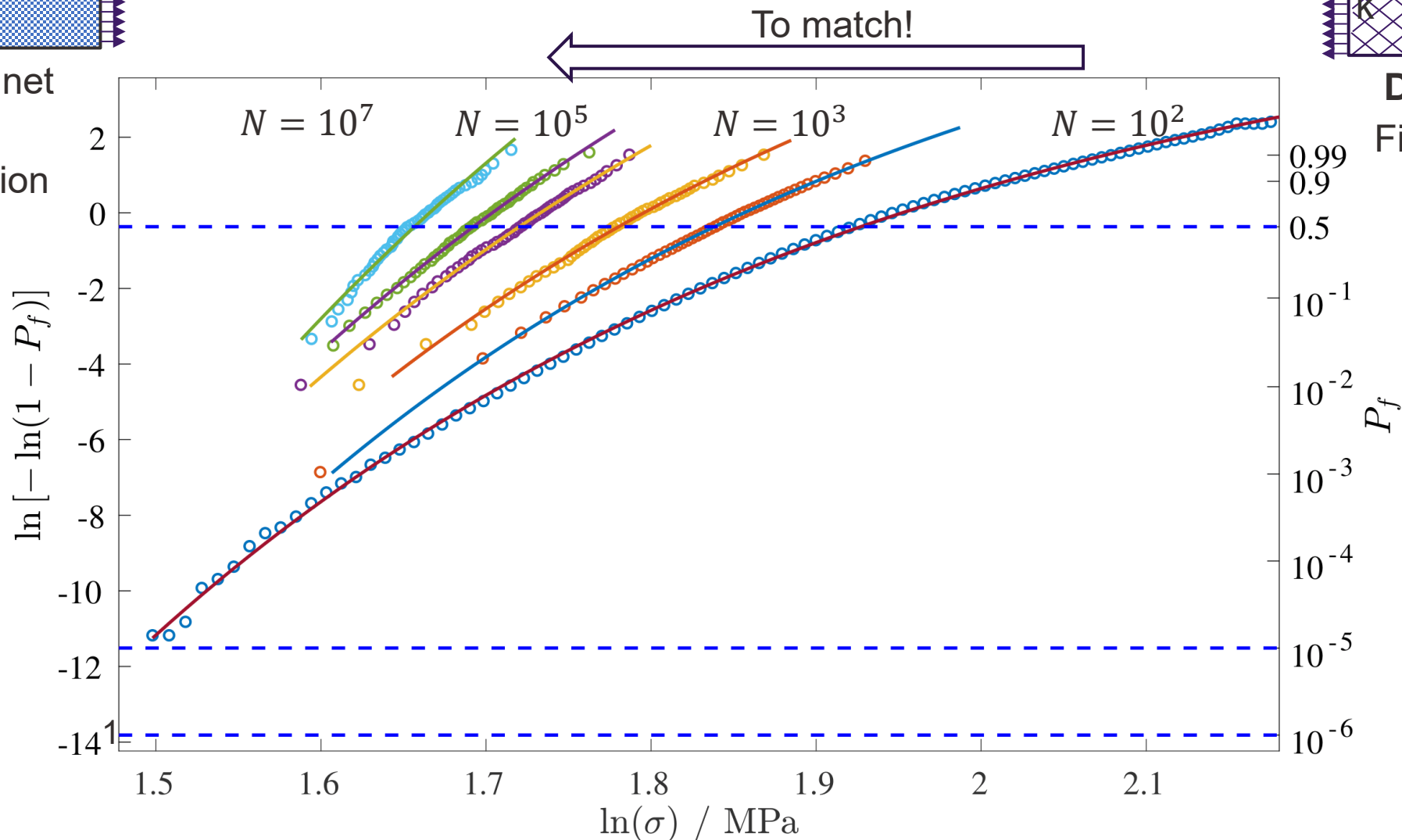
Discrete-to-Continuum Transition of Fishnet Statistics



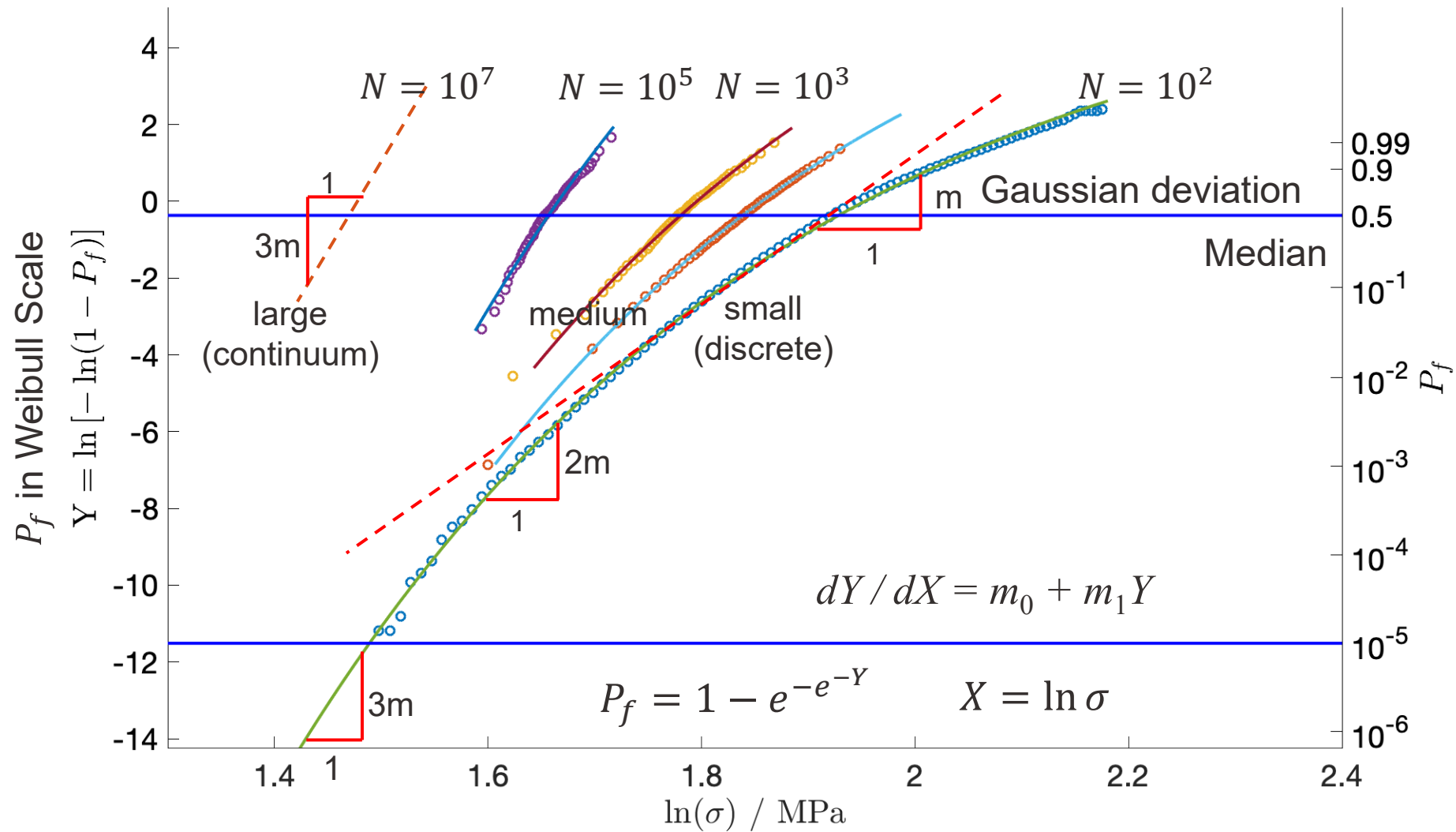
Continuum fishnet
(Laplace eq.)
Weibull distribution



Discrete fishnet
Fishnet distribution



Discrete-to-Continuum Transition of Fishnet Statistics



Conclusions

- A differential equation is derived for the evolution of pdf of fishnets of various sizes.
- Asymptotic matching of the failure distribution of small and large fishnets yields a general formula for estimating the fishnet failure probabilities up to 10^{-6} .
- Laplace equation for a continuum fishnet shows that a fully formed crack makes a large rectangular fishnet unstable.

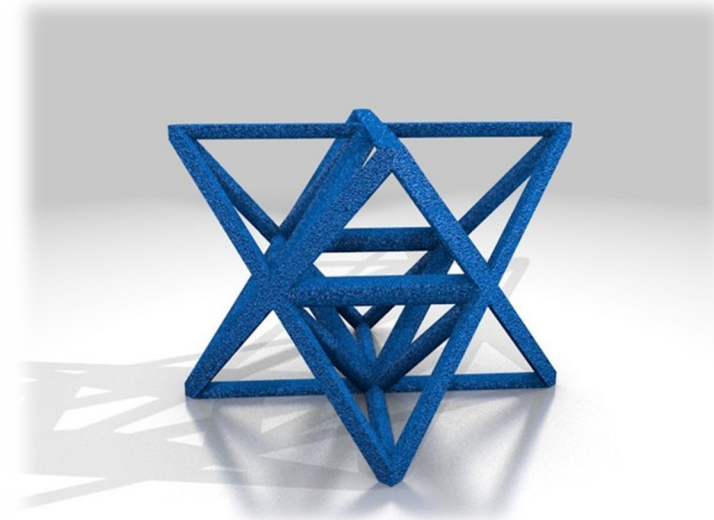
Future Work of Fishnet Statistics

Impact of Biaxial/Triaxial Tension on Failure Probability:

- Under biaxial/triaxial tension, failure probability may be altered. However, this adjustment does not change the distribution type because the topology of links will not change.

Challenges in Assessing Failure Probabilities for Complex Configurations:

- The introduction of irregular shapes, interconnected components through 3D printing, and hybrid configurations (parallel-series) adds complexity to the assessment of failure probabilities.
- The use of open-source codes, such as FEniCSx, could be helpful in HPC.





Thank you!

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